



*Pan-sharpen an image using PCA-based and IHS
algorithm*

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Introduction

Context

- - Growing need for high-resolution multispectral imagery in remote sensing
- - Challenge of balancing spatial and spectral resolution
- - Importance of pansharpening in modern remote sensing applications

Problem Statement

- - Need to enhance spatial resolution of multispectral imagery
- - Comparison of PCA and IHS approaches
- - Focus on implementation and quality assessment

Dataset Description

- - Landsat 8 OLI sensor data
- - Study area: New York region
- - Data specifications:
 - * Panchromatic band (15m resolution)
 - * Multispectral bands (30m resolution)

Theoretical Framework

Pansharpening Concept

- - Definition and purpose
- - General workflow:
 - * High-resolution panchromatic image
 - * Low-resolution multispectral image
 - * Fusion process
 - * Enhanced output

PCA Method Theory

- - Mathematical foundation
- - Principal component transformation
- - Component substitution process
- - Inverse transformation

IHS Method Theory

- - Color space transformation
- - Intensity component replacement
- - Back-transformation process
- - Color preservation considerations

Comparision

PCA Concept

Definition:

- Statistical approach transforming MS bands into uncorrelated components
- First principal component replaced with PAN image

Key Features:

- Handles unlimited spectral bands
- Better spectral preservation
- Computationally intensive
- Higher memory requirements
- Processing time: ~2-3x longer

Best Used For:

- Scientific analysis
- Complex terrain analysis
- Hyperspectral imagery
- When accuracy is critical

IHS Concept

Definition:

- Color space transformation technique
- Intensity component replaced with PAN image

Key Features:

- Limited to 3 bands (RGB)
- Faster processing
- Lower memory usage
- Simple implementation
- Good visual results

Best Used For:

- Quick visualization
- Urban mapping
- Real-time applications
- Limited computing resources

PCA-based Pansharpening Implementation ---

We have created a multispectral image using single R,G,B bands in the data

Detailed Workflow

Step-by-step process:

- 1. Data normalization
- 2. PCA transformation
- 3. Component substitution
- 4. Inverse transformation
- 5. Output scaling

Code Implementation

- Key algorithm components
- Memory optimization techniques
- Error handling strategies
- Output generation process

Processing Considerations

- Tile size optimization
- Computational efficiency and RAM limitations
- Quality control measures

IHS-based Pansharpening Implementation

Detailed Workflow

Step-by-step process:

- 1. RGB to IHS transformation
- 2. Intensity replacement
- 3. Back-transformation
- 4. Result optimization

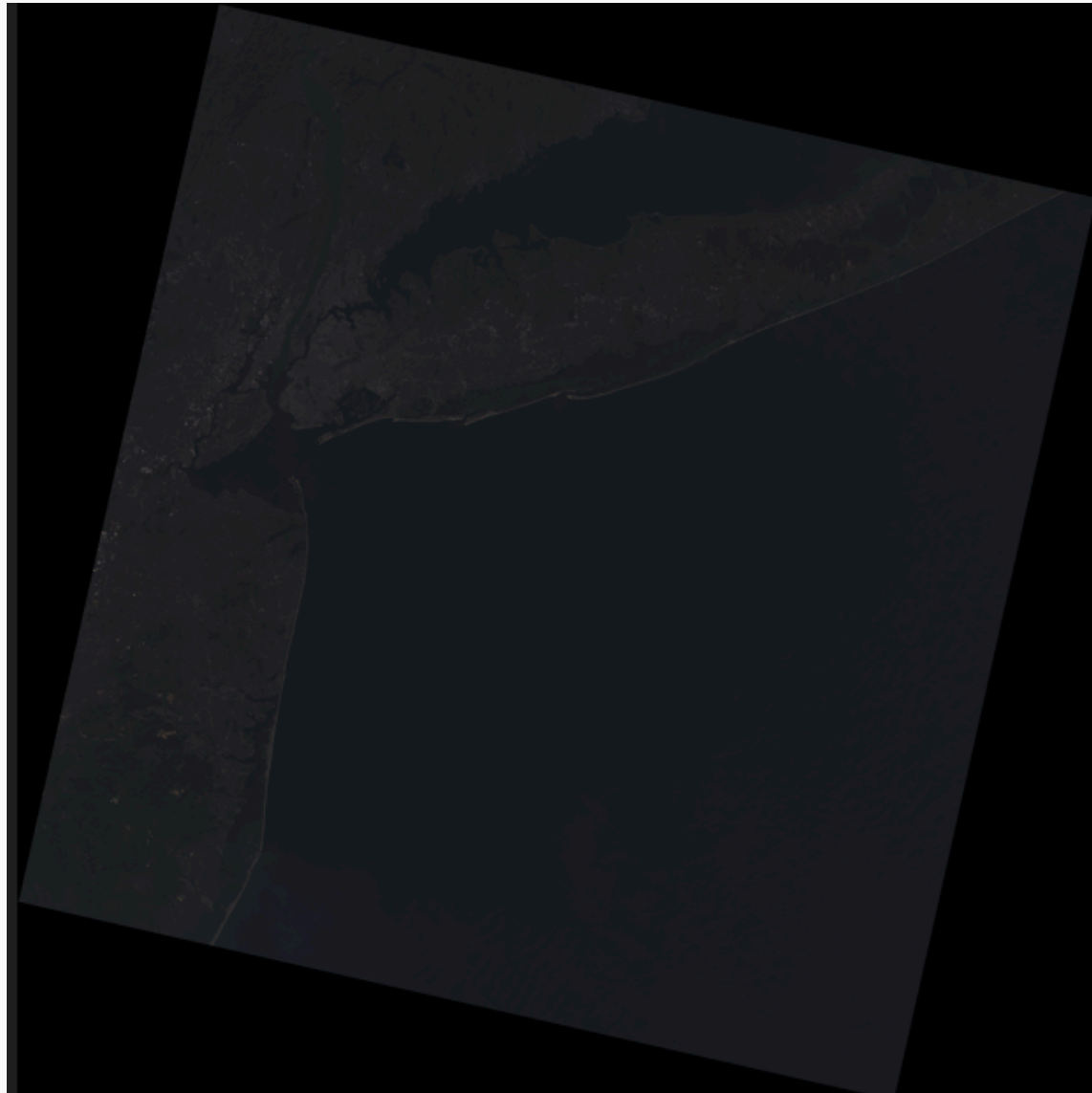
Code Implementation

- Transform equations
- Color space conversion
- Processing optimizations
- Output generation

Processing Considerations

- Color fidelity maintenance
- Processing speed optimization
- Error handling approaches

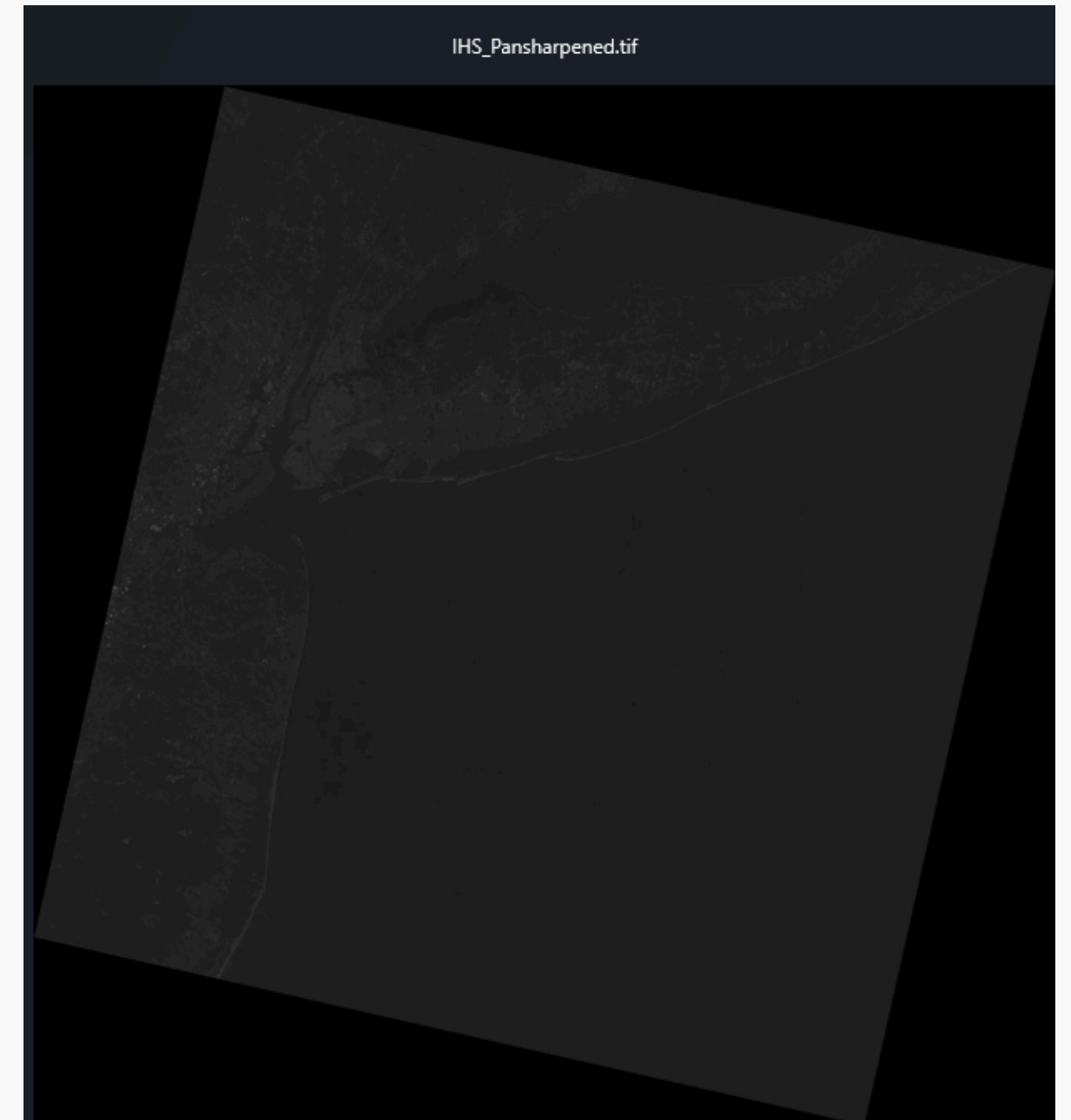
Results and Analysis



Input Pan Chromatic Image



PCA Based Pan Sharpening



IHS Based Pan Sharpening

Method Comparison

Spectral Quality ; Better for PCA based method

Spatial Quality; Better for PCA based method

Processing Speed: IHS>PCA

Memory Usage: PCA>IHS



Shortcomings:

- Extremely long run times
- Could only run PCA for cropped region, due to memory constraints





Thank you

